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Inventor(s)	Shaffer; Chadwick A. et al.

Walk outdoor power equipment unit having handle mounted operational controls using compliant mechanisms

Abstract

An outdoor power equipment unit, such as a lawn mower, includes a frame movable over the ground. A handle assembly is provided on the frame to allow an operator to walk on the ground behind the handle assembly while operating the unit. A traction drive and a ground grooming or working implement are carried on the frame. A traction control hand grip and an implement control bail are carried on the handle assembly. Both the hand grip and the bail are movable relative to the handle assembly by flexure provided by first and send compliant mechanisms.

Inventors: **Shaffer; Chadwick A. (Oakdale, MN), Wadzinski; Chris A. (Inver Grove Heights, MN), Weber; Jeff (Minneapolis, MN), Monitor; Shawn S. (Robbinsdale, MN)**

Applicant: **THE TORO COMPANY** (Bloomington, MN)

Family ID: **83689726**

Assignee: **THE TORO COMPANY (Bloomington, MN)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4503958	12/1984	Nishio	N/A	N/A
4667459	12/1986	Scanland	N/A	N/A
6945133	12/2004	Rush et al.	N/A	N/A
7762049	12/2009	Eaton	N/A	N/A
7762050	12/2009	Kaskawitz	N/A	N/A
2010/0025124	12/2009	Arpino	N/A	N/A
2010/0307908	12/2009	Selzer	N/A	N/A
2013/0046448	12/2012	Fan et al.	N/A	N/A
2016/0037719	12/2015	Bian	N/A	N/A
2017/0367259	12/2016	Shaffer	N/A	N/A
2021/0185912	12/2020	Wilks	N/A	N/A
2023/0098509	12/2022	Shaffer	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2805597	12/2013	EP	N/A
3329763	12/2017	EP	N/A
3471126	12/2018	EP	N/A
3666055	12/2019	EP	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability in PCT/US2022/043629, dated Jul. 12, 2023, 21 pages. cited by applicant

International Search Report and Written Opinion in PCT/US2022/043629, dated Dec. 14, 2022, 12 pages. cited by applicant

Written Opinion of the International Preliminary Examining Authority in PCT/US2022/043629, dated Aug. 8, 2023, 9 pages. cited by applicant

Operator's Manual, “Self Propelled Mower, Rear Wheel Drive Model Series Q and R,” Form No. 769-18650A, Troy-Bilt LLC, Valley City, OH, USA, dated 2019 [retrieved Apr. 18, 2024]. Retrieved from the Internet: <URL: https://www.troybilt.com/en_US/self-propelled-mowers/tbwc23b-xp-self-propelled-lawn-mower/12ABR27B766.html>, 28 pages. cited by applicant

Parts Diagram, “Troy-Bilt Model TBWC23 XP (2021) > Control Bails.” Troy-Bilt LLC, Valley City, OH, USA, [retrieved Apr. 18, 2024]. Retrieved from the Internet: <URL: https://www.troybilt.com/en_US/ari-partstream-host.html>, 2 pages. cited by applicant

European Examination Report issued Apr. 25, 2025, for European Patent Application No. 22787051.6, 8 pages. cited by applicant

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Primary Examiner: Risic; Abigail A

Attorney, Agent or Firm: Mueting Raasch Group

Background/Summary

(1) This application is a continuation application of U.S. patent application Ser. No. 17/488,643, filed Sep. 29, 2021, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD
(1) This invention relates to a walk outdoor power equipment unit, such as a walk power lawn mower, having various powered components which are controlled by an operator using operational controls mounted on a handle of the unit.
BACKGROUND OF THE INVENTION
(2) Walk power mowers of the type commonly used by homeowners to cut the grass in their yards are well known. Such mowers have a frame that carries various wheels for allowing movement of the frame over the ground. The frame is shaped to provide a cutting deck having a downwardly facing cutting chamber that houses one or more rotary blades for cutting grass. The frame may have a traction drive to self-propel the frame over the ground.
(3) A power source, such as an internal combustion engine or an electric motor powered by an onboard battery pack, is mounted on the frame atop the cutting deck. The power source is suitably coupled to whatever cutting blades are enclosed in the cutting chamber to rotate such cutting blades in horizontal cutting planes. The power source is also suitably coupled to the traction drive to rotate or power at least some of the wheels carried on the frame. The traction drive relieves the operator of having to manually propel the frame over the ground.
(4) The frame of the mower includes an upwardly and rearwardly extending handle to allow the operator to guide the frame during a grass cutting operation and to manually propel the frame in a mower which lacks a traction drive. Various hand operated operational controls are typically located on the upper end of the handle. Such controls allow the operator to engage and disengage the rotation of the blades and/or to engage and disengage the operation of the traction drive. When the traction drive is capable of propelling the mower at a variable ground speed, such controls will also allow the operator to set or maintain the ground speed desired by the operator.
(5) The operational controls are often mounted on the handle for pivoting about lateral, substantially horizontal pivot axes. This requires the use of various pivots, bushings and fasteners for securing the controls to the handle while providing smooth and reliable pivoting of the controls on the handle. In addition, various springs are often required as well to normally dispose the controls in a disengaged condition or in a neutral position when the controls are bi-directionally movable. While the use of such mechanical components is effective for the purpose of mounting the controls on the handle and for providing the motion required of the controls, such components add complexity and cost to the mower.
SUMMARY OF THE INVENTION
(6) One aspect of this invention relates to a walk outdoor power equipment unit for performing a ground grooming or working operation. The unit comprises a frame which is movable over the ground. A prime mover is carried by the frame. A traction drive is carried by the frame and is operatively powered by the prime mover for self-propelling the frame over the ground at a variable speed. An implement is carried by the frame and is operatively powered by the prime mover for

performing the ground grooming or working operation. A handle assembly is carried by on the frame and is configured for use by an operator who walks on the ground. At least one control is carried by the handle assembly for controlling at least one of the traction drive or the implement. The at least one control acts through a compliant mechanism to effect control of the at least one of the traction drive or implement.

(7) One aspect of this invention relates to a walk outdoor power equipment unit for performing a ground grooming or working operation. The unit comprises a frame which is movable over the ground. A prime mover is carried by the frame. A traction drive is carried by the frame and is operatively powered by the prime mover for self-propelling the frame over the ground at a variable speed. An implement is carried by the frame and is operatively powered by the prime mover for performing the ground grooming or working operation. A handle assembly is carried by on the frame and is configured for use by an operator who walks on the ground. At least one control is carried by the handle assembly for controlling at least one of the traction drive or the implement. The at least one control is movably mounted on the handle assembly by a compliant mechanism.

(8) Another aspect of this invention relates to an outdoor power equipment unit. The unit comprises a frame movable over the ground. A handle assembly is provided on the frame to allow an operator to walk on the ground behind the handle assembly while operating the unit. A traction drive and a ground grooming or working implement are carried on the frame. A traction control hand grip and an implement control bail are carried on the handle assembly. The hand grip and the bail are movable relative to the handle assembly by flexure provided by a pair of compliant mechanisms, respectively.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This invention will be described in detail hereafter in the Detailed Description, when taken in conjunction with the following drawings, in which like reference numerals refer to like elements throughout.

(2) FIG. **1** is a perspective view of one example, i.e., a walk power mower, of a walk outdoor power equipment unit equipped with the operational controls of a first embodiment of this invention mounted to an upper portion of a handle of the unit;

(3) FIG. **2** is an enlarged perspective view of the operational controls of FIG. **1**;

(4) FIG. **3** is a partial perspective view of the operational controls of FIG. **1**, the controls being shown in exploded form for the sake of clarity;

(5) FIG. **4** is an enlarged perspective view of a first compliant mechanism used to mount one of the operational controls of FIG. **1**, i.e., a traction control, to the handle of the walk outdoor power equipment unit, particularly illustrating a magnet and Hall sensor incorporated in the first compliant mechanism;

(6) FIG. **5** is a side elevational view of the first compliant mechanism for pivotally mounting the traction control to the handle of the walk outdoor power equipment unit;

(7) FIG. **6** is a side elevational view of a second compliant mechanism which pivotally mounts an implement control directly to a portion of the first compliant mechanism as a way of indirectly mounting the implement control to the handle of the walk outdoor power equipment unit;

(8) FIG. **7** is an enlarged perspective view of a second embodiment of the operational controls according to this invention, the first mechanism for mounting the traction control to the handle no longer comprising a compliant mechanism but now comprising a first non-compliant mechanism having a rigid pivot pin and for producing pivotal motion of the traction control and a mechanical spring for biasing the traction control towards neutral;

(9) FIG. **8** is an exploded view of the second embodiment of the operational controls of FIG. **7**,

particularly illustrating the first non-complaint mechanism in an exploded form for the sake of clarity;
(10) FIG. **9** is an enlarged perspective view of a third embodiment of the operational controls according to this invention, the first and second mechanisms for mounting the traction control and the implement control the handle no longer comprising compliant mechanisms but now comprising first and second non-compliant mechanisms each having a rigid pivot pin and a mechanical spring with the implement control acting through a compliant mechanism to engage and disengage the operation of the implement; and
(11) FIG. **10** is an exploded view of the third embodiment of the operational controls of FIG. **9**, particularly illustrating the compliant mechanism and the first and second non-complaint mechanisms that pivotally mount the traction and implement controls to the handle.
DETAILED DESCRIPTION
(12) One example of a walk outdoor power equipment unit **2** having the operational controls of this invention is illustrated in FIG. **1**. In this example, walk outdoor power equipment unit **2** comprises a walk power mower **4** of the type homeowners commonly use to cut the grass in their lawns. However, the operational controls of this invention are not limited for use on a mower **4**, but could be used on other walk outdoor power equipment units **2** having a powered implement for performing a ground grooming or working operation and/or a traction drive for self-propelling unit **2**. These units **2** could comprise a snowblower, an aerator, a trencher, a stump grinder, a blower, and the like.
(13) Referring further to mower **4** of FIG. **1** by way of example, mower **4** has a frame **6** supported for movement over the ground by a plurality of wheels **8**. A central portion of frame **6** forms a generally circular cutting deck **10** which provides a downwardly facing cutting chamber **12** that is open at the bottom. A prime mover **14**, e.g. an internal combustion engine, an electric motor powered by an onboard battery pack, or the like, is mounted atop cutting deck **10**. A vertical drive shaft (not shown) of prime mover **14** extends downwardly into cutting chamber **12**.
(14) In mower **4** of FIG. **1**, at least one grass cutting blade (not shown) is coupled in any suitable manner to the drive shaft of prime mover **14**. The blade comprises or the blades collectively comprise the powered implement for performing a ground grooming or working operation. In this case, that operation is cutting grass. The results of that operation, namely the grass clippings generated by the operation of the blade(s), are thrown into a rearwardly disposed bag **16** in a collecting mode of operation or may be driven downwardly into the cut swath in a mulching mode of operation if bag **16** is not installed on mower **4** or mower **4** was not built to accommodate bag **16**.
(15) In one embodiment of this invention, mower **4** has a traction drive (not shown) for self-propelling frame **6** of mower **4** over the ground at least in a forward direction at a variable ground speed. In other embodiments, the traction drive could be configured to self-propel frame **6** in both forward and reverse directions at variable ground speeds in each direction. The variable ground speeds could be infinitely variable, e.g. by infinitely changing the speed of an electric motor driving an axle that mounts rear wheels **8** or by shifting a continuously variable transmission driving the axle. Alternatively, the variable ground speeds could be changed in discreet steps, e.g. by shifting a multi-speed transmission driving the axle.
(16) As further shown in FIG. **1**, mower **4** includes an upwardly and rearwardly extending handle assembly **18**. Handle assembly **18** includes a pair of parallel, laterally spaced, side tubes **20** connected at their lower ends to frame **6** of mower **4**. Side tubes **20** are joined together at their upper ends by a substantially horizontal cross tube **22**. When the traction drive is not powered, the operator would be able to manually maneuver frame **6** of mower **4** by gripping cross tube **22** and by using cross tube **22** to push forwardly or pull rearwardly on frame **6** of mower **4**.
(17) Handle assembly **18** includes a traction speed control that in one embodiment thereof comprises a laterally extending, pivotal hand grip **24** carried on the upper rear end of handle

assembly **18** forward of and above cross tube **22** of handle assembly **18**. During an actual grass cutting operation, the operator's hands will usually be gripping hand grip **24** rather than cross tube **22**. Handle assembly **18** further includes an implement control that in one embodiment thereof disengages and engages the operation of the blade(s) of mower **4**. As shown in FIG. **1**, the implement control may comprise a laterally extending, pivotal bail **26** having a general size and shape that mimics that of hand grip **24**. Bail **26** may be positioned forward of and slightly below hand grip **24**. Other positions of bail **26** relative to hand grip **24** are possible.

(18) The blade(s) of mower **4** are disengaged when bail **26** is spaced away from hand grip **24** as shown in FIG. **1**. The blade(s) of mower **4** are engaged when the operator reaches out with the fingers of at least one hand and pivots bail **26** into a closed position closer to or against hand grip **24**. This blade engagement continues for so long as the operator holds bail **26** in the closed position. Bail **26** will automatically return to its spaced position relative to hand grip **24** to disengage the blade(s) when the operator releases bail **26**. If so desired, some type of bail latch (not shown) that must be released could be required in order for the operator to close bail **26** against hand grip **24** to mitigate the chance of the operator inadvertently starting the operation of the blade(s).

(19) In some embodiments, this invention relates to the use of one or more compliant mechanisms to mount either or both of traction control hand grip **24** and implement control bail **26** on the top of handle assembly **18**. In mechanical engineering, a compliant mechanism is commonly defined as a flexible mechanism that achieves force and motion transmission through elastic body deformation. It gains some or all of its motion from the relative flexibility of its members rather than from rigid-body joints alone. The term compliant mechanism shall be used herein in accordance with this definition.

(20) Referring now to FIGS. **2-6**, a first compliant mechanism **28** is duplicated on each side of handle assembly **18** to affix the opposite ends of hand grip **24** to the upper ends of side tubes **20** of handle assembly **18**. As best shown in FIGS. **2** and **3**, first compliant mechanism **28** has a fixed body portion **30** substantially rigidly secured to the top of the upper end of side tube **20** by a plurality of threaded fasteners **32**. Fasteners **32** extend upwardly through apertures in the bottom of side tube **20** and thread into internally threaded bores contained in posts **34**, best seen in FIG. **5**, provided on fixed body portion **30** of first compliant mechanism **28**. When each first compliant mechanism **28** is being installed on its respective side tube **20**, posts **34** are pushed into the interior of side tube **20** through apertures in the top of side tube **20** with the threaded bores of posts **34** aligning with the apertures in the bottom of side tube **20**. When fasteners **32** are then pushed up through the apertures in the bottom of side tube **20** and are then tightened in the downwardly facing bores of posts **34**, fixed body portion **30** of each first compliant mechanism **28** will be substantially rigidly mounted to its respective side tube **20**.

(21) First compliant mechanism **28** further includes a movable body portion **36** that is an integral part of first compliant mechanism **28** but is able to flex in a generally pivotal manner with respect to fixed body portion **30**. Referring to FIG. **3**, movable body portion **36** has an upwardly facing, generally cylindrical socket **38** that receives the downwardly facing, generally cylindrical end **40** of one side of hand grip **24**. Once hand grip end **40** is inserted into socket **38**, a laterally extending, horizontal bolt **42** passes through aligned apertures in hand grip end **40** and socket **38**. When a nut **44** and washer **46** are tightened on the end of bolt **42**, hand grip end **40** is substantially affixed to movable body portion **36** of first compliant mechanism **28**.

(22) In the embodiment shown in FIG. **3**, bolt **42** does double duty as it extends beyond first compliant mechanism **28** to rigidly affix a portion of a second compliant mechanism **48** to the inner side of first compliant mechanism **28**. Second compliant mechanism **48** movably mounts bail **26** adjacent hand grip **24** in a manner that will be described more fully hereafter. However, in embodiments of the invention in which the implement control formed by bail **26** is not used and there is no second compliant mechanism **48**, bolt **42** shown in FIG. **3** will be shorter and the nut **44**

and washer **46** will directly engage against the inner side of first compliant mechanism **28** rather than the inner side of second compliant mechanism **48**.

(23) Referring now to FIG. **5** hereof, the centerline **39** of socket **38** of movable body portion **36** of first compliant mechanism **28** is aligned with a narrow web **50** which connects the movable and fixed body portions **28**, **48** of first compliant mechanism **28** together to unify them into a unitary structure. A first relatively long, open ended, front slot **52** is placed forward of web **50** and extends through the thickness of first compliant mechanism **28**. A second relatively short, open ended, rear slot **54** is placed rearward of web **50** and also extends through the thickness of first compliant mechanism **28**. Second slot **54** has an open end that comprises a relatively small slit **56** while the first slot has an open end that is much larger.

(24) Referring to FIG. **3**, movable body portion **36** of first compliant mechanism **28** has a front end **58**. The opposite sides of front end **58** include two reduced diameter portions **60** that extend laterally outwardly from opposite sides of front end **58**. Reduced diameter portion **60** on the inward side of front end **58** of movable body portion **36** of first compliant mechanism **28** will be received within a front circular opening **62** on second compliant mechanism **48**. This helps locate second compliant mechanism **48** on first compliant mechanism **28**.

(25) As best shown in FIG. **4**, front end **58** of movable body portion **36** of first compliant mechanism **28** has a vertical bore **64** extending downwardly from the top thereof. A small cylindrical magnet **66** can be inserted into and pushed downwardly into bore **64** until magnet **66** rests atop an inwardly extending ledge (not shown) within bore **64** as shown in FIG. **4**. Magnet **66** can be retained in this position by a pair of flexible detents (not shown) on opposite sides of bore **64** which allow magnet **66** to pass as it is inserted through bore **64** but which then snap over on top of magnet **66** once magnet **66** has reached the ledge. Other ways of attaching magnet **66** to front end **58** of movable body portion **36** of first compliant mechanism **28** could be used.

(26) Referring now to FIGS. **3** and **4**, the top of a front end of fixed body portion **30** of first compliant mechanism **28** includes an upwardly open pocket **68**. An electrically operated sensor, such as, but not limited to, a Hall sensor **70**, is installed in a horizontal slideway **72** at the top of pocket **68** as best shown in FIG. **4**. Two apertures **74** also shown in FIG. **4** are disposed respectively on either side of slideway **72**. Once Hall sensor **70** is fully inserted in slideway **72**, two screws **76** are threaded into apertures **74**. When screws **76** are so installed, portions of the heads of screws **76** will abut against opposite sides of Hall sensor **70** to retain Hall sensor **70** in slideway **72** as best shown in FIG. **3**.

(27) FIG. **5** illustrates first compliant mechanism **28** in a neutral position in which it has not been deformed or flexed by force applied to hand grip **24**. If the operator wishes to adjust the traction speed of mower **4** and is gripping hand grip **24**, the operator need only push forwardly on hand grip **24**. This will cause front end **58** of movable body portion **36** of first compliant mechanism **28** to flex or pivot generally downwardly towards fixed body portion **30** in the direction of arrow A in FIG. **5**. This motion is permitted by front slot **52** in first compliant mechanism **28** which will narrow somewhat as hand grip **24** is pushed forwardly and by rear slot **54** which will expand somewhat as hand grip **24** is pushed forwardly.

(28) Conversely, the operator cannot meaningfully move hand grip **24** in the opposite or rearward direction from neutral. This is prevented by the very narrow slit **56** that forms the opening on rear slot **54**. Slit **56** will substantially immediately closed upon such rearward motion of hand grip **24**. Thus, in a unidirectional variable speed traction drive embodiment of the invention under consideration here, movable body portion **36** of first compliant mechanism **28** becomes rigid almost immediately upon any motion of hand grip **24** in a rearward direction from neutral. In other bidirectional, variable speed traction embodiments, slit **56** that forms the open end of rear slot **54** could be much wider to allow movable body portion **36** of first compliant mechanism **28** to flex meaningfully rearwardly when hand grip **24** is moved rearwardly out of neutral.

(29) Together, magnet **66** and Hall sensor **70** provide a motion sensing device that can infinitely

determine the degree of motion of movable body portion **36** relative to fixed body portion **30**. This motion is caused by the operator moving hand grip **24** forwardly to set or select a desired forward traction speed. The motion information provided by Hall sensor **70** is sent to an electronic controller on mower **4**. The electronic controller then varies the operational speed of the traction drive in concert with how far hand grip **24** has been pivoted forwardly by the operator out of a neutral position thereof.

(30) Referring now to FIGS. **3** and **6**, second compliant mechanism **48** has a fixed body portion **78** that includes a lateral passage **80** that receives bolt **42** that secures movable body portion **36** of first compliant mechanism **28** to hand grip **24**. In addition, reduced diameter portion **60** of front end **58** of movable body portion **36** of first compliant mechanism **28** is inserted into circular opening **62** of fixed body portion **78** of second compliant mechanism **48**. When nut **44** and washer **46** are tightened on bolt **42**, fixed body portion **78** of second compliant mechanism **48** will be drawn towards and clamped tightly against the inner side of movable body portion **36** of first compliant mechanism **28** as best shown in FIG. **2**. Thus, as movable body portion **36** of first compliant mechanism **28** flexes during motion of hand grip **24** to control the traction drive, second compliant mechanism **48** will move with movable body portion **36** of first compliant mechanism **28** with bail **26** maintaining a fixed orientation relative to hand grip **24**.

(31) Second compliant mechanism **48** has a movable body portion **82** which includes an upwardly facing, generally cylindrical socket **84** that receives the downwardly facing, generally cylindrical end **86** of one side of bail **26**. Once bail end **86** is inserted into bail socket **84**, a laterally extending fastener (not shown) passes through aligned apertures in bail end **86** and bail socket **84** to secure bail **26** to movable body portion **82** of second compliant mechanism **48**. When the pair of second compliant mechanisms **48** are mounted on the pair of first compliant mechanism **28** and the opposite ends **86** of bail **26** are inserted into bail sockets **84** on the pair of laterally spaced second compliant mechanisms **48**, bail **26** will be disposed in an implement disengaged position spaced away from hand grip **24** as best shown in FIG. **2**.

(32) Referring now to FIG. **6** hereof, the centerline **85** of bail socket **84** of movable body portion **82** of second compliant mechanism **48** is aligned with a narrow web **88** that connects the fixed and movable body portions **78**, **82** of second compliant mechanism **48** together to unify them into a unitary structure. A first relatively long, open ended, front slot **90** is placed forward of web **88** and extends through the thickness **82** of second compliant mechanism **48**. A second relatively long, open ended, rear slot **92** is placed rearward of web **88** and also extends through the thickness of first compliant mechanism **28**. First slot **90** has an open end that comprises a relatively small slit **94** while second slot **92** has an open end that is much larger.

(33) As further shown in FIG. **6**, fixed body portion **78** of second compliant mechanism **48** includes an upwardly open switch housing **96** that receives the case of an on/off type electrical switch **98**. Switch **98** has a plunger **100** that is outwardly biased by an internal spring within the case. Plunger **100** when fully extended from the case in the off position of switch **98** is in contact with or spaced close to a rear wall **102** of movable body portion **82** of second compliant mechanism **48**. In order to change the state of switch **98** from off to on, the rear wall of movable body portion **82** of second compliant mechanism **48** must be moved rearwardly in FIG. **6** to depress plunger **100** relative to the case of switch **98** sufficiently to change the state of switch **98** to on. When switch **98** is placed into the on state, the controller of mower **4** will receive an activation signal and will engage the operation of the implement, e.g. start the blade(s) of mower **4** rotating when outdoor power equipment unit **2** comprises a mower **4** as shown in FIG. **1**.

(34) The motion of movable body portion **82** of second compliant mechanism **48** which is needed to actuate switch **98** is caused by the operator closing bail **26** from its usual position spaced away from hand grip **24** to an implement engaged position adjacent to or in contact with hand grip **24**. The force applied by bail **26** to movable body portion **82** of second compliant mechanism **48** is sufficient to flex movable body portion **82** of second compliant mechanism **48** about web **88**

relative to fixed body portion **78** of second compliant mechanism **48**. This will cause rear wall **102** of movable body portion **82** of second compliant mechanism **48** to flex or pivot generally downwardly/rearwardly towards switch **98** in the direction of arrow B in FIG. **6**. This motion is permitted by rear slot **92** in second compliant mechanism **48** which will narrow somewhat as bail **26** is pulled rearwardly and by front slot **90** which will expand somewhat as bail **26** is pulled rearwardly. The amount of motion allowed upon the closing of bail **26** will be sufficient to change the state of switch **98** from off to on to thereby start the operation of the implement.

(35) One material which can be used to manufacture first compliant mechanism **28** and/or second compliant mechanism **48** or any other compliant mechanism described in this application is a thermoplastic polyester elastomer such as DuPont Hytrel 7246.

(36) Each compliant mechanism **28**, **48** could be used in an outdoor power equipment unit **2** without using the other. For example, in another embodiment of the invention depicted in FIGS. **7** and **8** hereof, first compliant mechanism **28** has been removed in favor of a first non-compliant mechanism **28′** for mounting hand grip **24** to handle. Each first non-compliant mechanism **28′** is duplicated on both side tubes **20** when hand grip **24** extends across the full width of handle **18**. Since first non-compliant mechanism **28′** is similar to first compliant mechanism **28**, the reference numerals used to refer to the same components of first compliant mechanism **28** will be used to refer to the corresponding components of first non-compliant mechanism **28′** with a single prime designation following the reference numeral.

(37) First non-compliant mechanism **28′** bas a substantially rigid fixed body portion **30′** and a substantially rigid movable body portion **36′** having overall shapes and functions similar to their compliant counterparts **30**, **36**. As best shown in FIG. **8**, fixed body portion **30′** has a cavity **104** that includes aligned apertures **106** in spaced side walls **108** of cavity **104**. A protrusion or plug **110** on an underside of a rear portion of movable body portion **36′** fits down into cavity **104**. Plug **110** has a through bore **112** that is aligned with apertures **106** in side walls **108** of cavity **104** when movable body portion **36′** is assembled onto fixed body portion **30′**. A rigid pivot pin **114** then passes through apertures **106** and bore **112**. Pivot pin **114** can be secured by a bolt (not shown) on one end of pivot pin **114** to provide a mechanical pivot connection between fixed body portion **30′** and movable body portion **36′**.

(38) As shown in FIG. **7** and in addition to pivot pin **114**, a mechanical compression spring **116** normally biases movable body portion **36′** in a direction that places hand grip **24** in a neutral position in which the speed of the traction drive is zero. The neutral position could be established by a physical stop (not shown) between fixed body portion **30′** and movable body portion **36′** that limits further pivotal motion of hand grip **24** rearwardly.

(39) When the operator pushes forwardly on hand grip **24** to control the speed of the traction drive, the movable body portion **36′** will pivot downwardly about pivot pin **114** to compress spring **116** against fixed body portion **30′**. Magnet **66** and Hall sensor **70** are present in this embodiment and will function as in the first embodiment of FIGS. **2-6** to control the traction speed depending upon the degree of downward pivoting of movable body portion **36′** relative to fixed body portion **30′**. When the operator releases hand grip **24**, the compression built up in spring **116** will return hand grip **24** to its neutral position.

(40) Second compliant mechanism **48** as disclosed in FIGS. **1-6** is still used in the embodiment of FIGS. **7** and **8** to support bail **26** for pivotal motion relative to hand grip **24**. Second compliant mechanism **48** is clamped against the side of movable body portion **36′** of first non-compliant mechanism **28′** by bolt **42** which passes through aperture **80** in second compliant mechanism **48**. In addition, the opening **62** in the nose of second compliant mechanism **48** is still received on the reduced diameter portion **60′** (hidden in FIG. **8**) on the inward side of front end **58′** of movable body portion **36′** of first non-compliant mechanism **28′**. This helps prevent any rotation of second compliant mechanism **48** relative to first non-compliant mechanism **28′** about the axis of bolt **42**. Second compliant mechanism **48** as used in the embodiment of FIGS. **7** and **8** works identically to

its operation in the embodiment of FIGS. **1-6**.
(41) In a further variation, first compliant mechanism **28** could be used with second compliant mechanism **48** being removed and being replaced by a non-compliant mechanism which includes a rigid pivot and mechanical spring.
(42) Referring now to FIG. **9**, an additional embodiment of this invention comprises a first non-compliant mechanism **28**″ which is duplicated and mounted on each side tube **20** of handle assembly **18**. First non-compliant mechanism **28**″ of FIGS. **9** and **10** is a modified version of the first non-compliant mechanism **28**′ of FIGS. **7** and **8**. Since first non-compliant mechanism **28**″ is similar to first non-compliant mechanism **28**′, the reference numerals used to refer to the same components of first compliant mechanism **28**′ will be used to refer to the corresponding components of first non-compliant mechanism **28**″ with a double prime designation following the reference numerals.
(43) In this embodiment, there are no second compliant mechanisms **48** for providing pivotal motion of bail **26**. Instead and referring to FIG. **10**, the bottom of each side of bail **26** has an outwardly turned circular end that forms a rigid pivot pin **118**. Each pivot pin **118** is rotatably received in a circular aperture of bushing **120** on movable body portion **36**″ of each first non-compliant mechanism **28**″. This pivotally journals the opposite ends of bail **26** directly on the pair of laterally spaced first non-compliant mechanisms **28**″, **28**″ mounted on side tubes **20**, **20** of handle **18**.
(44) As further shown in FIG. **10**, a coil **122** of a torsion spring **124** is concentrically installed around a pivot pin **118**. One leg **126** of torsion spring **124** is hooked around a rear portion of one side of bail **26**. The other leg **128** of torsion spring **124** will abut against a fixed portion of movable body portion **36**″ of the adjacent first non-compliant mechanism **28**″ to which the one side of bail **26** is journalled.
(45) Normally, the bias of spring **124** will serve to move bail **26** to its neutral position spaced away from hand grip **24** as shown in FIG. **9**. When bail **26** is pivoted to its closed position adjacent to or in contact with hand grip **24**, torsion spring **124** will be wound up to store force therein. When the operator releases bail **26**, this stored up force in torsion spring **124** will return bail **2** to its neutral position. Torsion spring **124** can be used on just one pivot pin **118** on one side of bail **26** or can be duplicated and used on both pivot pins **118** on both sides of bail **26**.
(46) In the embodiment of FIGS. **9** and **10**, the second compliant mechanism **48** disclosed in FIGS. **1-8** is no longer needed to pivotally mount bail **26** in view of the mounting(s) provided by pivot pin **118** and spring **124**. However, a modified second compliant mechanism **48**″ is still used with respect to the operation of bail **26**. Since second compliant mechanism **48**″ is similar to second compliant mechanism **48** of FIGS. **1-8**, the reference numerals used to refer to the same components of second compliant mechanism **48** will be used to refer to the corresponding components of second compliant mechanism **48**″ with a double prime designation following the reference numerals.
(47) As shown in FIG. **9**, second compliant mechanism **48**″ comprises a fixed body portion **78**″ that is clamped to the inward side of movable body portion **36**″ of first non-compliant mechanism **28**″. Referring to FIG. **10**, this is accomplished by bolt **42** which passes through an aperture **80**″ in second compliant mechanism **48**″ to pull fixed body portion **78**″ of second complaint mechanism **48**″ tightly against first non-compliant mechanism **28**″ when a nut (not shown) on the free end of bolt **42** is tightened. In addition, the opposite sides of a rear end **130** of movable body portion **36**″ of first non-compliant mechanism **28**″ include two reduced diameter portions **60**″ that extend laterally outwardly from opposite sides of rear end **130**. The reduced diameter portion **60**″ on the inward side of rear end **130** (not visible in FIG. **10**) will be received within a rear circular opening **62**″ on second compliant mechanism **48**″. The engagement of reduced diameter portion **60**″ in opening **62**″ helps prevent any rotation of second compliant mechanism **48**″ relative to first non-compliant mechanism **28**″ about the axis of bolt **42**.

(48) Fixed body portion **78″** of second compliant mechanism **48″** still includes a switch housing **96″** that receives the case of an on/off type electrical switch **98″**. Switch **98″** has a plunger **100″** that is outwardly biased by an internal spring within the case. Second compliant mechanism **48″** has a movable body portion **82″** that can flex relative to fixed body portion **78″** by virtue of an elongated slot **92″** that permits such flexure. A rear wall **102″** of movable body portion **82″** is slightly spaced from or in slight contact with plunger **100″** of switch **98″**. Second compliant mechanism **48″** is used on only one side of bail **26** as it not needed on the other side of bail **26**.
(49) As shown in FIGS. **9** and **10**, when bail **26** is in its neutral or implement disengaged condition, bail **26** is spaced forwardly of movable body portion **82″** of second compliant mechanism **48″**. The upper portion of movable body portion **82″** includes a forwardly facing U-shaped cradle **132** which is shaped to receive a rearward portion of one leg **134** of bail **26**. When the operator pivots bail **26** from its neutral position towards hand grip **24**, the first lost motion portion of its travel takes up the space between bail **26** and cradle **132**. Once bail leg **134** seats itself within cradle **132**, further motion of bail **26** towards hand grip **24** causes movable body portion **82″** of second compliant mechanism **48″** to flex or pivot generally downwardly/rearwardly towards switch **98** in the direction of arrow B″ in FIG. **10**. The amount of such flexure is sufficient to depress plunger **100″** and thereby actuate switch **98″** to begin the operation of the powered implement, e.g. the blades of mower **4**.
(50) The use of either or both of the first and second compliant mechanisms **28**, **48** to mount traction drive and implement controls on the handle of a walk outdoor power equipment unit **2** is an advance in the art. It avoids the cost of using numerous rigid pivots and mechanical springs in the limited spaces available for the controls at the top of handle assembly **18**. In addition, while the embodiment of FIGS. **9** and **10** use rigid pivots and mechanical springs to mount the traction drive and implement controls, it still retains the use of a compliant mechanism **48″** arranged in a lost motion manner relative to bail **26** to operate switch **98″**. This allows bail **26** to have a larger throw or arc of movement relative to hand grip **24**, i.e. bail **26** when its neutral position is spaced further away from hand grip **24**, as shown by comparing the positons of bail **26** relative to hand grip **24** as shown in FIGS. **2** and **9**.
(51) Various modifications of this invention will be apparent to those skilled in the art. For example, the use of a hand grip **24** and bail **26** that span the full width of handle assembly **18** and which are mounted by duplicate first and second compliant mechanisms to side tubes **20** is only one preferred embodiment of such controls. Hand grip **24** and bail **26** could be mounted only to one side tube **20** and extend only partially across the width of handle assembly **18**.
(52) Various other modifications will be apparent to those skilled in the art. Accordingly, the scope of this invention is to be limited only by the appended claims.

Claims

1. A power equipment unit comprising: a frame movable over a ground surface; one or more prime movers carried by the frame; a handle assembly carried by the frame and configured for use by an operator; a traction drive carried by the frame and operatively powered by one of the one or more prime movers for self-propelling the frame over the ground surface at a variable ground speed; an implement carried by the frame and operatively powered by one of the one or more prime movers for performing a ground grooming or working operation; a compliant mechanism comprising: a fixed body portion operatively coupled to the handle assembly; and a movable body portion coupled to the fixed body portion; and a traction control and an implement control connected to the movable body portion, wherein each of the traction control and the implement control are movable relative to the handle assembly by flexure provided by the compliant mechanism.
2. The power equipment unit of claim 1, wherein the compliant mechanism comprises: a first compliant mechanism carrying the traction control; and a second compliant mechanism carrying

the implement control.

3. The power equipment unit of claim 2, wherein the first compliant mechanism comprises: a fixed body portion secured to the handle assembly; a movable body portion; and a web integrally connecting the fixed body portion to the movable body portion.

4. The power equipment unit of claim 3, wherein the second compliant mechanism is mounted to the first compliant mechanism.

5. The power equipment unit of claim 3, wherein the second compliant mechanism comprises: a fixed body portion secured to the movable body portion of the first compliant mechanism; a movable body portion; and a web integrally connecting the fixed body portion of the second compliant mechanism to the movable body portion of the second compliant mechanism.

6. The power equipment unit of claim 1, wherein the traction control comprises a hand grip.

7. The power equipment unit of claim 1, wherein the implement control comprises a bail.

8. The power equipment unit of claim 1, wherein the compliant mechanism comprises a thermoplastic polyester elastomer.

9. The power equipment unit of claim 1, wherein the handle assembly comprises a first side tube and a second side tube, and wherein the compliant mechanism includes a first compliant mechanism coupled to the handle assembly at or near the first side tube, and a second compliant mechanism coupled to the handle assembly at or near the second side tube.

10. The power equipment unit of claim 1, further comprising a magnet secured to one of the movable body portion and the fixed body portion of the compliant mechanism, and a Hall sensor secured to the other of the movable body portion and the fixed body portion of the compliant mechanism, wherein the Hall sensor is configured to detect movement of the magnet during movement of the movable body portion relative to the fixed body portion.

11. The power equipment unit of claim 1, further comprising a non-compliant mechanism comprising: a fixed body portion secured to the handle assembly; and a movable body portion pivotally connected to the fixed body portion, wherein the fixed body portion of the compliant mechanism is connected to the movable body portion of the non-compliant mechanism, and the movable body portion of the compliant mechanism is connected to the fixed body portion of the compliant mechanism by a flexure.

12. The power equipment unit of claim 11, further comprising a switch secured to the fixed body portion of the compliant mechanism, the switch comprising a plunger selectively actuated by the movable body portion of the compliant mechanism.

13. A power equipment unit for performing a ground grooming or working operation, the unit comprising: a frame movable over a ground surface; one or more prime movers carried by the frame; a traction drive carried by the frame and operatively powered by one of the one or more prime movers for self-propelling the frame over the ground surface at a variable ground speed; an implement carried by the frame and operatively powered by one of the one or more prime movers for performing the ground grooming or working operation; a handle assembly carried by the frame; a traction drive control operatively coupled to the handle assembly by a traction drive control mechanism, wherein the traction drive control mechanism comprises: a fixed body portion secured to the handle assembly; and a movable body portion movable relative to the fixed body portion to support the traction drive control; and an implement control operatively coupled to the handle assembly by an implement control mechanism, wherein the implement control mechanism comprises: a fixed body portion secured to the movable body portion of the traction drive control mechanism; and a movable body portion movable relative to the fixed body portion of the implement control mechanism to support the implement control.

14. The power equipment unit of claim 13, wherein the fixed body portion of the traction drive control mechanism is connected to the movable body portion of the traction drive control mechanism by a flexure.

15. The power equipment unit of claim 13, wherein the fixed body portion of the implement control

mechanism is connected to the movable body portion of the traction drive control mechanism, and wherein the fixed body portion of the implement control mechanism is connected to the movable body portion of the implement control mechanism by a flexure.